

- 3 Fig. 3.1 shows an aircraft flying at constant speed in a horizontal circle of radius r . The weight of the aircraft is W and L is the lift force acting at right angles to the wings.

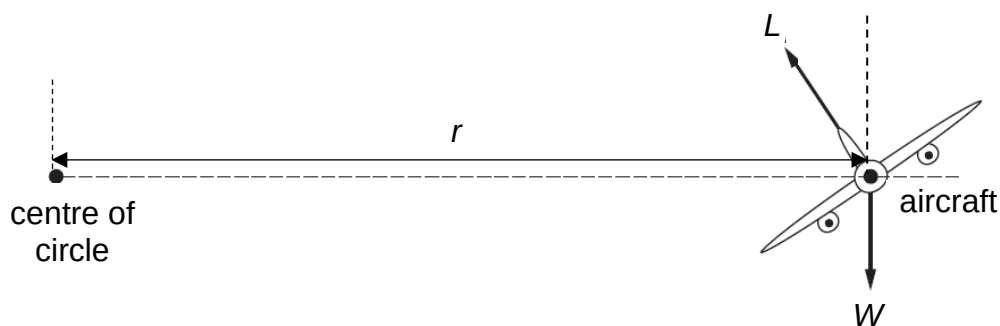


Fig. 3.1

- (a) Explain how the lift force L maintains the aircraft flying in a horizontal circle.

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[2]

- (a) **Solution:**
The vertical component of L balances the weight W of aircraft.
The horizontal component of L provides the centripetal force for the aircraft to fly in a horizontal circle.

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- (b) The weight W of the aircraft is 3.92×10^5 N. With the aircraft flying at a constant speed of 250 m s^{-1} in a horizontal circle, the lift force L acts at angle of 35° to the vertical.

- (i) State the vertical component of L .

vertical component of $L = \dots\dots\dots$ N

[1]

- (i) **Solution:**
Vertical component of $L = 3.92 \times 10^5$ N

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- (ii) Calculate L .

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A 500 g mass is attached to a spring on a horizontal frictionless surface, and is set in